LOAN APPROVAL PREDICTION -POWER BI DASHBOARD

To analyze customer demographic details such as age, gender, city, education, and marital status.

To examine the distribution of customers based on employment type and annual income.

To analyze loan amount, loan term, and credit score across different customer groups.

To identify the relationship between credit score and loan approval status.

To compare loan approval and rejection rates across cities and employment types.

To visualize key financial indicators such as total loan amount, average income, and average credit score.

To identify high-risk customer segments using credit score and loan details.

To build an interactive Power BI dashboard for loan approval analysis and decision making.

Dataset Description

This dataset contains information about 500 customers and their loan-related details. It includes both numerical and categorical variables, which are used to analyze customer profiles, loan behavior, and loan approval status.

Data Cleaning & Preprocessing

The dataset was checked for missing values, duplicate records, and incorrect data types before visualization. Since the dataset contains both numerical and categorical columns, proper data cleaning was performed to ensure accurate analysis in Power BI.

Steps performed

Checked the dataset for missing or null values in all 12 columns.

Removed duplicate customer records, if any, using CustomerID.

Verified that numerical columns such as Age, AnnualIncome, LoanAmount, CreditScore, and LoanTermMonths contain valid numeric values.

Checked categorical columns such as Gender, City, Education, MaritalStatus, EmploymentType, and LoanStatus for spelling errors or inconsistent values.

Confirmed that LoanStatus contains only two valid categories:

Approved and Rejected.

Converted incorrect data types into the correct format, such as text, whole number, and decimal number.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the overall structure, patterns, and relationships in the loan dataset. Power BI visuals were used to analyze customer demographics, financial details, and loan approval status.

Key analysis performed

Analyzed the total number of customers, total loan amount, and average credit score.

Examined the distribution of customers by gender, city, education, and marital status.

Compared annual income across different employment types.

Analyzed loan amount distribution based on city and employment type.

Studied the relationship between credit score and loan approval status.

Identified approval and rejection rates across customer groups.

Compared average credit score for approved and rejected loans.

Analyzed loan terms and their impact on loan approval.

Key insights

Customers with higher credit scores generally have a higher chance of loan approval.

Salaried and business customers show different income and loan amount patterns.

Loan approval and rejection rates vary across cities and employment types.

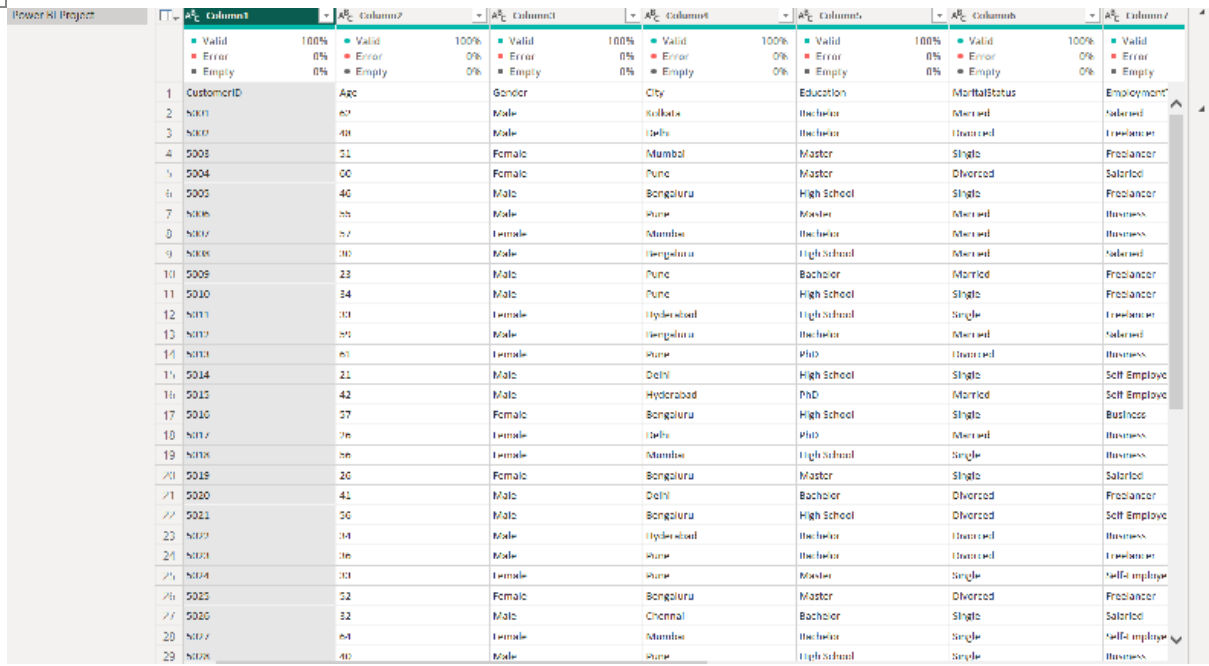
Annual income and loan amount are important factors for understanding customer financial behavior.

Credit score is a major indicator used to assess loan risk.

Created calculated measures in Power BI for total loan amount, average credit score, and loan approval rate.

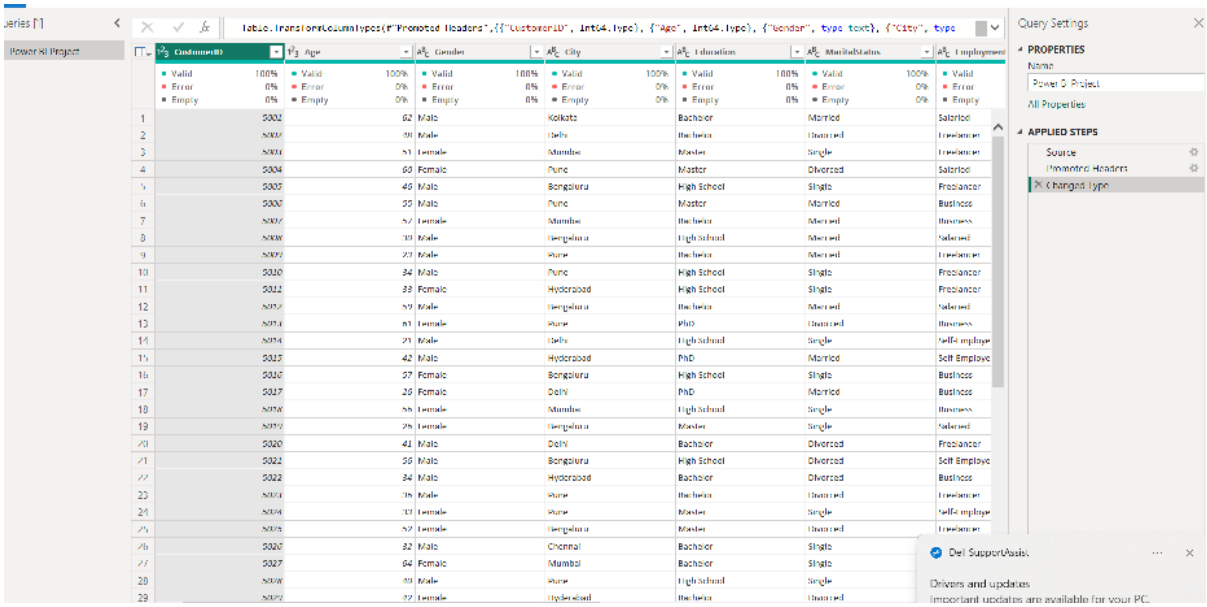
Prepared the cleaned dataset for creating interactive dashboards and visual reports

IMPORTING THE DATA:



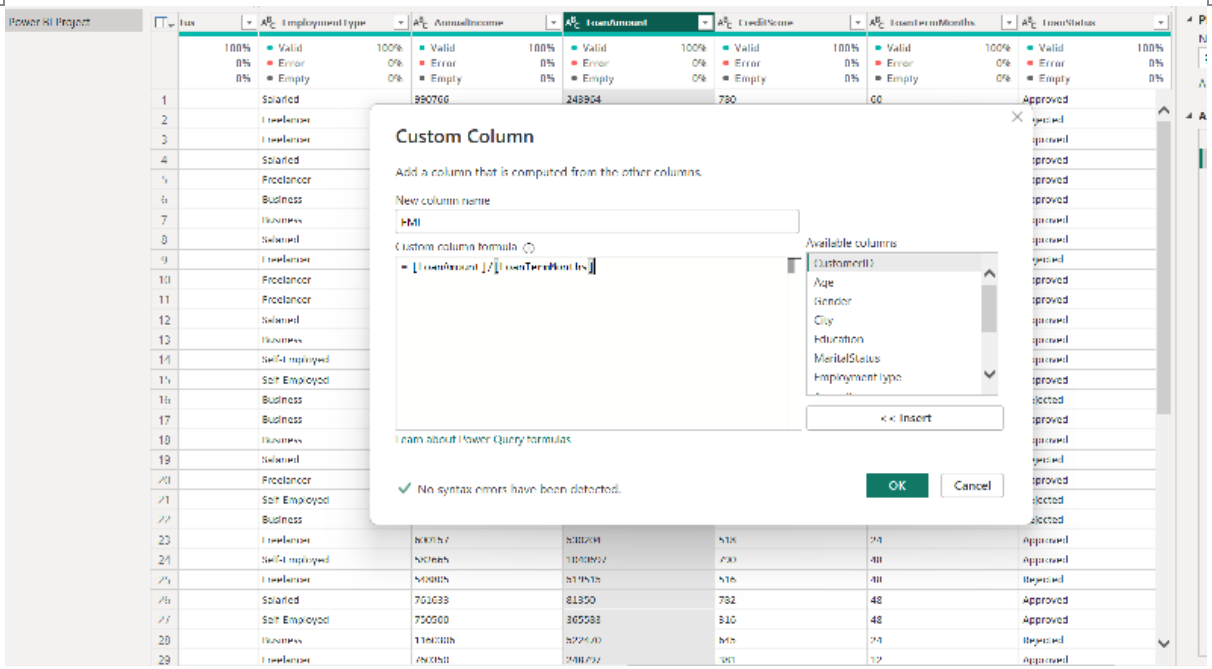
	CustomerID	Age	Gender	City	Education	MaritalStatus	Employment
1	CustomerID	Age	Gender	City	Education	MaritalStatus	Employment
2	5001	62	Male	Kolkata	Bachelor	Married	Salaried
3	5002	48	Male	Delhi	Bachelor	Divorced	Freelancer
4	5003	31	Female	Mumbai	Master	Single	Freelancer
5	5004	60	Female	Pune	Master	Divorced	Salaried
6	5005	46	Male	Bangaluru	High School	Single	Freelancer
7	5006	55	Male	Pune	Master	Married	Business
8	5007	57	Female	Mumbai	Bachelor	Married	Business
9	5008	30	Male	Bangaluru	High School	Married	Salaried
10	5009	23	Male	Pune	Bachelor	Married	Freelancer
11	5010	34	Male	Pune	High School	Single	Freelancer
12	5011	33	Female	Hyderabad	High School	Single	Freelancer
13	5012	52	Male	Bangaluru	Bachelor	Married	Salaried
14	5013	61	Female	Pune	PhD	Divorced	Business
15	5014	21	Male	Delhi	High School	Single	Self Employe
16	5015	42	Male	Hyderabad	PhD	Married	Self Employe
17	5016	57	Female	Bangaluru	High School	Single	Business
18	5017	26	Female	Delhi	PhD	Married	Business
19	5018	55	Female	Mumbai	High School	Single	Business
20	5019	20	Female	Bangaluru	Master	Single	Salaried
21	5020	41	Male	Delhi	Bachelor	Divorced	Freelancer
22	5021	56	Male	Bangaluru	High School	Divorced	Self Employe
23	5022	34	Male	Hyderabad	Bachelor	Divorced	Business
24	5023	26	Male	Pune	Bachelor	Divorced	Freelancer
25	5024	33	Female	Pune	Master	Single	Self-Employe
26	5025	52	Female	Bangaluru	Master	Divorced	Freelancer
27	5026	32	Male	Chennai	Bachelor	Single	Salaried
28	5027	64	Female	Mumbai	Bachelor	Single	Self-Employe
29	5028	40	Male	Pune	High School	Single	Business

REMOVING DUPLICATES:

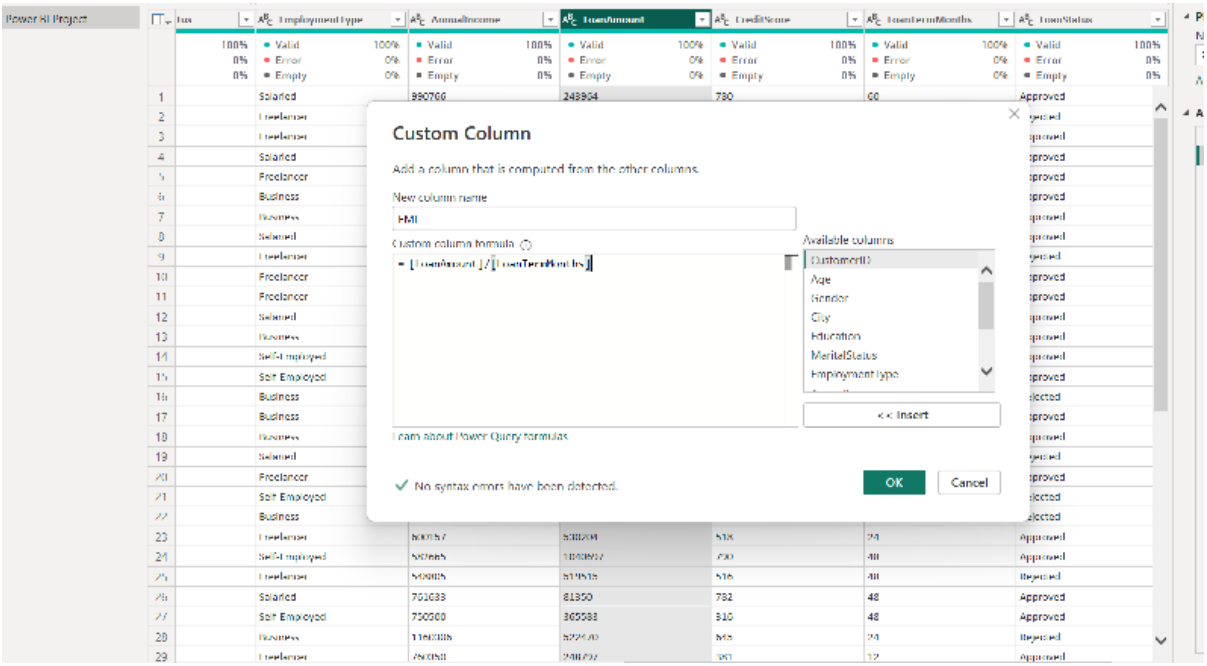


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FILTERING AND SORTING:



CREATING NEW COLUMN AS EMI:

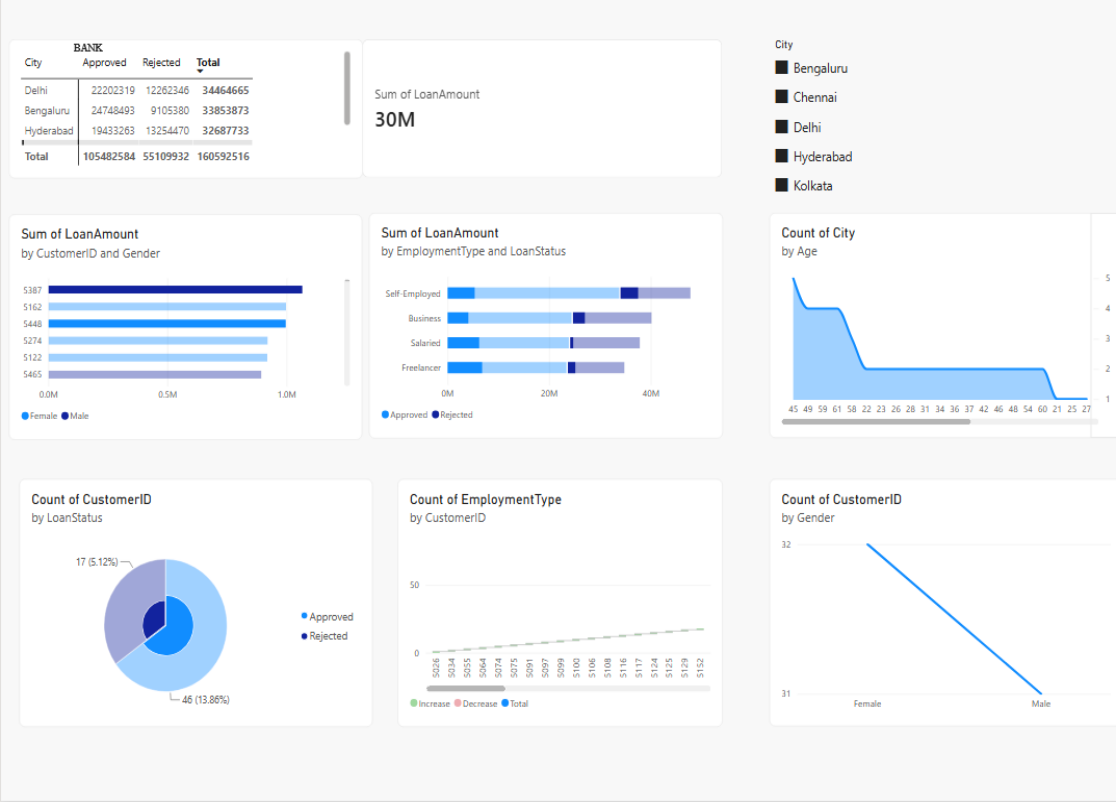


DATA VISUALIZATION USING POWER BI:

Multiple visualizations that communicate insights clearly

Using of charts(BAR,LINE,PIE,SLICER,CARDS

Proper titles and labels



Description

This project presents an interactive Power BI dashboard developed using a loan dataset containing 500 customer records and 12 attributes. The dataset includes customer demographic details, financial information, loan details, and loan approval status.

The main objective of this report is to analyze customer and loan-related data and identify the key factors that influence loan approval decisions. Power BI visuals were used to analyze customer age, gender, city, education, marital status, employment type, annual income, loan amount, credit score, and loan term.

The dashboard provides a clear view of total customers, total loan amount, average credit score, and loan approval rate. It also compares loan approval and rejection rates across different cities, employment types, and customer groups. The analysis shows that credit score, annual income, and loan amount are important factors in understanding loan eligibility and customer risk.

Overall, this Power BI report helps convert raw loan data into meaningful visual insights. It supports better decision-making by identifying customer patterns, high-risk segments, and important factors affecting loan approval. The dashboard can be used by banks or financial institutions to analyze loan applications and make data-driven lending decisions.

Prepared By

Name: Lavanya
Project Title: Loan Approval Prediction – Power BI Dashboard
Dataset: Loan Approval Dataset (500 Records)
Tool Used: Microsoft Power BI